

Definition 0.1 (*f*-Divergence): Let P and Q be probability measures on a measurable space $(\mathcal{X}, \mathcal{F})$. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a convex function with $f(1) = 0$, extended to $[0, \infty]$ by lower semicontinuity.

Let $P = P_{\text{ac}} + P_{\text{s}}$ be the Lebesgue decomposition of P with respect to Q . The f -divergence of P from Q is defined as

$$D_f(P\|Q) = \int_{\mathcal{X}} f\left(\frac{dP_{\text{ac}}}{dQ}\right) dQ + f'(\infty) P_{\text{s}}(\mathcal{X}),$$

where $f'(\infty) = \lim_{t \rightarrow \infty} \frac{f(t)}{t}$ represents the recession constant of f .

If $P \ll Q$, then $P_{\text{s}} = 0$ and the definition simplifies to:

$$D_f(P\|Q) = \int_{\mathcal{X}} f\left(\frac{dP}{dQ}\right) dQ.$$

Remark 0.2: The definition of f -divergence is independent of any choice of densities or coordinates on $\mathcal{X}$. All expressions are formulated directly in terms of Radon–Nikodym derivatives and Lebesgue decompositions, making the definition valid for discrete, continuous, and mixed distributions.

Remark 0.3: The singular component P_{s} captures the portion of P supported on sets of Q -measure zero. The term $f'(\infty) P_{\text{s}}(\mathcal{X})$ represents the information-theoretic cost of this singular mismatch. For many divergences of interest (e.g. Kullback–Leibler), $f'(\infty) = \infty$, implying $D_f(P\|Q) = \infty$ whenever $P \not\ll Q$.

Remark 0.4: When $P \ll Q$, the f -divergence reduces to an expectation under Q of a convex function of the likelihood ratio $\frac{dP}{dQ}$. This highlights the role of convexity in establishing properties such as non-negativity, lower semicontinuity, and the data processing inequality.

Remark 0.5: Many classical information measures arise as special cases of f -divergences. For example, the Kullback–Leibler divergence corresponds to $f(t) = t \log t$, and the total variation distance corresponds to $f(t) = \frac{1}{2}|t - 1|$.

0.0.1 See also

0.0.2 References